Unit Plan Summary

Objectives:

- Students will use Finite Element Analysis (FEA) to evaluate the structural integrity of a bridge truss under various load conditions
- Students will use MatLab and ABAQUS software to calculate nodal displacement and element stress

Week 1: Vector and Matrix Quantities

- Demonstrate how matrices can be used to solve a system of two linear equations with unknowns
- Identify an inverse of a matrix and finding the determinant
- Solve a system of three linear equations with three unknowns using matrices
- Understand the fundamental rules of matrices and matrix multiplication

Common Core State Standards:

- **CCSS.MATH.CONTENT.HSN-VM.C.10 (+)**
  Understand that the zero and identity matrices play a role in matrix addition and multiplication similar to the role of 0 and 1 in the real numbers. The determinant of a square matrix is nonzero if and only if the matrix has a multiplicative inverse.
- **CCSS.MATH.CONTENT.HSN-VM.C.11 (+)**
  Multiply a vector (regarded as a matrix with one column) by a matrix of suitable dimensions to produce another vector. Work with matrices as transformations of vectors.
- **CCSS.MATH.CONTENT.HSN-VM.C.12 (+)**
  Work with 2×2 matrices as transformations of the plane, and interpret the absolute value of the determinant in terms of area.

Week 2: Introduction to FEA and Hooke's Law

- Introduction to the fundamentals of FEA, computational mesh components such as nodes and elements.
- Introduction to Hooke's Law and using F = k x to evaluate stiffness, displacement and force on the load.
- Derive the formula for a 1 - D spring element and expending the formula for multiple element system

NGSS STANDARD: HS - PS2 - 6

- Communicate scientific and technical information about how the underlying molecular and atomic-level structures decide the bulk properties of materials.

  Hooke's Law (F=kx), material elasticity vs. plastic deformation, and element stiffness matrix assembly ([K]{u}={F}) to show how individual element properties dictate macroscopic structural behavior.

Week 3: Math Lab Geometry and Unit Cell construction

- Place nodes in the coordinate plane and connect them using elements
- Construct a 5 x 5 Unit Cell with the diagonals using nodes and elements
- Construct a trapezoidal bridge truss using nodes and elements
- Construct a 5 x 5 Unit Cell with the diagonals using nested loops.

NGSS Standard: HS-ETS1-2 Engineering Design

- Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

NGSS Standard: HS-ETS1-4 Engineering Design

- Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems.

Week 4:

- Introduce ABAQUS core features such as constraints and construction tools. Use such tools to construct a trapezoidal truss feature from Week 3
- Translate MatLab file into an input file and import an input file into ABAQUS
- Apply loads to the structure and set boundary conditions.

NGSS Standard: HS-ETS1-4: Engineering Design

- Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems.